

EXPLORING PANSORI GENERATION WITH ACE-STEP

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ABSTRACT

Pansori is a traditional Korean narrative musical form that combines sung passages with spoken narration (*aniri*) under explicit rhythmic cycles (*jangdan*). While recent text-to-music models achieve strong results on mainstream genres, their behavior on underrepresented traditions remains largely unexplored. We adapt ACE-Step to pansori via LoRA-based fine-tuning and compare its adaptation to Korean pop as a mainstream baseline. We evaluate our results through quantitative analysis with Fréchet Audio Distance (FAD) and qualitative assessment by an expert performer. Our study highlights challenges and partial successes in adapting to pansori, and suggests directions for extending text-to-music models toward cultural diversity and traditional music.

1. INTRODUCTION

Recent advances in text-to-music models [1–3] from commercial services demonstrated that these models are capable of generating plausible music with proper styles in audio, including vocal with lyric condition. Following these services, open source models that can generate whole song audio have been introduced [4–7]. While these models show strong performance in generating popular music genres, their usability for generating traditional music of various countries is still under-explored [8,9].

In this work, we investigate the adaptation of ACE-Step [7] to *pansori*. *Pansori* [10] is a traditional Korean narrative musical form with sparse drum accompaniment, cyclical rhythmic frameworks (*jangdan*), and spoken passages (*aniri*). Its structure does not follow the verse–chorus patterns typical of popular music. These properties make *pansori* fundamentally different from mainstream music. This study explores the capabilities and limitations of adapting ACE-Step to *pansori*, and discusses their future directions for text-to-music models applied to traditional genres across cultures.

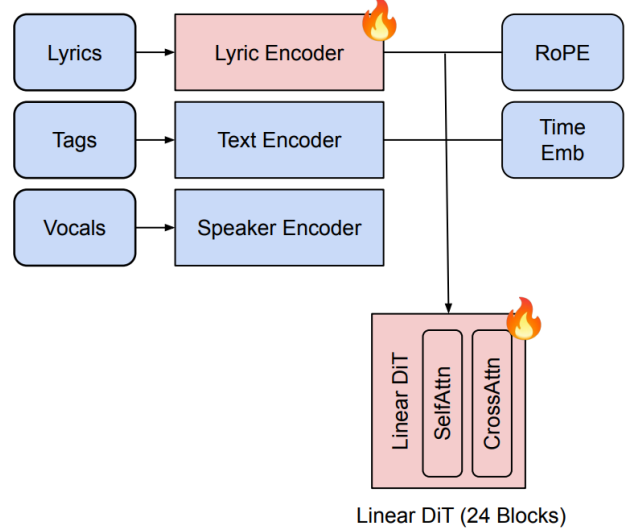


Figure 1. ACE-Step Overview

2. METHOD

2.1 Dataset

We used 7 hours of *pansori* audio performed by Jueun Lee, a lead artist of Folk Music Group of National Gugak Center, with permission from the performer. In this work, we define *segment markers* as symbolic labels that include *jangdan*, rhythmic patterns performed on a drum when singing, and *aniri*, spoken narrative passages rather than sung parts. These markers were enclosed in square brackets in the lyrics to provide explicit cues for the model. Audio was segmented into clips along lyric lines. We ensured that each *segment marker* was preserved and followed ACE-Step’s recommended maximum audio length of 4 minutes per clip. For comparison, we included 2 hours of Korean pop songs performed by Kwang-seok Kim.

2.2 LoRA Fine-tuning

The model was trained for 23,000 steps using LoRA with the default configuration (rank $r = 256$ and $\alpha = 32$). As illustrated in Figure 1, LoRA is applied to the lyric encoder and the main transformer.



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LoRA weights	Training steps				
	5k	10k	15k	20k	23k
1	0.929	0.771	0.821	0.812	0.643
2	0.601	0.481	0.430	0.544	0.423
3	0.655	0.626	0.596	0.765	0.648

Table 1. FAD↓ scores for pansori across LoRA weights and training steps.

LoRA weights	Training steps				
	5k	10k	15k	20k	23k
1	0.439	0.398	0.395	0.435	0.396
2	0.484	0.598	0.543	0.571	0.595
3	1.170	1.303	1.257	1.378	1.463

Table 2. FAD↓ scores for Korean pop across LoRA weights and training steps.

3. RESULTS

3.1 Analysis between Pansori and Korean pop

To quantitatively compare the model’s adaptation to pansori and Korean pop, we computed the Fréchet Audio Distance (FAD) [11] scores for each genre under different LoRA weights (rows in Tables) and training steps (columns in Tables). The results are summarized in Table 1 and Table 2.

For pansori, Table 1 shows that the lowest FAD is 0.423 at 23k steps with LoRA 2. LoRA 3 outperforms LoRA 1 up to 20k. For Korean pop, Table 2 indicates that lower FAD values generally occur with LoRA 1 and early checkpoints, while LoRA 3 yields higher FAD across steps.

Overall, pansori benefits from sufficient LoRA weights and training steps, whereas Korean pop tends to perform best with minimal adaptation. While Korean pop outperformed pansori in FAD scores at the lowest LoRA weight, its performance degraded significantly when the LoRA strength was increased.

3.2 Expert Feedback

To assess the quality of pansori generation, we consulted Jueun Lee, the original performer of the dataset and a professional gugak musician. According to her evaluation, the model captured vocal timbre, expressive ornamentation, and contextually appropriate segmentation when symbolic markers like *jangdan* and *aniri* were provided.

Although *jangdan* may sound sparse compared to pop drum accompaniment, it follows a well-defined rhythmic framework that the model did not successfully learn. However, the model often failed to maintain *jangdan* cycles, resulting in unstable timing and unclear rhythmic phrasing.

The model also struggled with narrative segments *aniri*. While the resulting *aniri* passages were distinct from sung segments, they still felt awkward and unnatural, failing to reflect their theatrical and speech-like nature. We provide audio examples on our supplementary webpage.¹

¹ We plan to release a comparison page with reference and generated

4. CONCLUSION

FAD results suggest that the further a genre is from the model’s pretraining distribution, the harder it is to steer generation through fine-tuning.

According to expert evaluation, the model captured short-term characteristics of pansori—such as vocal texture, ornamentation, and segment transitions—but failed to reproduce long-term characteristics of pansori—such as melodic progression and rhythmic cycles. *Aniri* lacked the natural theatrical quality expected of spoken narrative. These issues reflect the model’s difficulty in handling the mixed characteristics of pansori, which includes both singing and speech-like narration. Because the model is singing-oriented generation framework.

These findings imply that adaptation to traditional genres like pansori demands more than just parameter tuning. Rhythm-aware modeling, symbolic guidance, or architectural modifications may be necessary for more faithful generation.

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